

Chapter 6 outline

6.1 Introduction

Wireless

- ❑ 6.2 Wireless links, characteristics
 - CDMA
- ❑ 6.3 IEEE 802.11 wireless LANs ("wi-fi")
- ❑ 6.4 Cellular Internet Access
 - architecture
 - standards (e.g., GSM)

Mobility

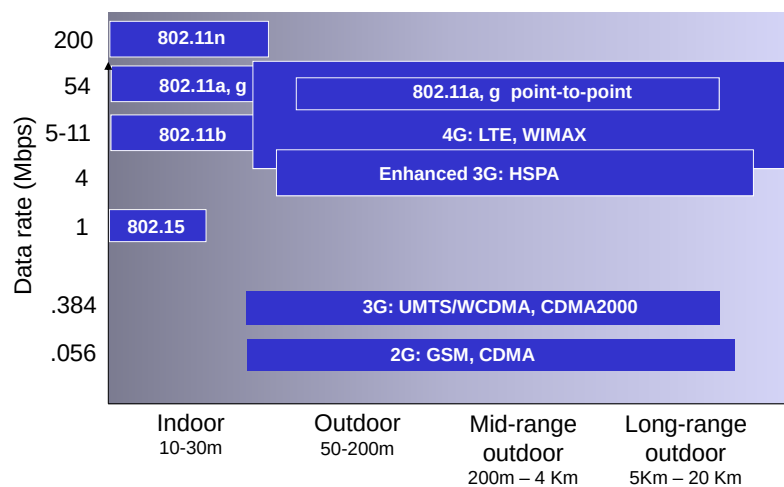
- ❑ 6.5 Principles: addressing and routing to mobile users
- ❑ 6.6 Mobile IP
- ❑ 6.7 Handling mobility in cellular networks
- ❑ 6.8 Mobility and higher-layer protocols

6.9 Summary

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Characteristics of selected wireless link standards



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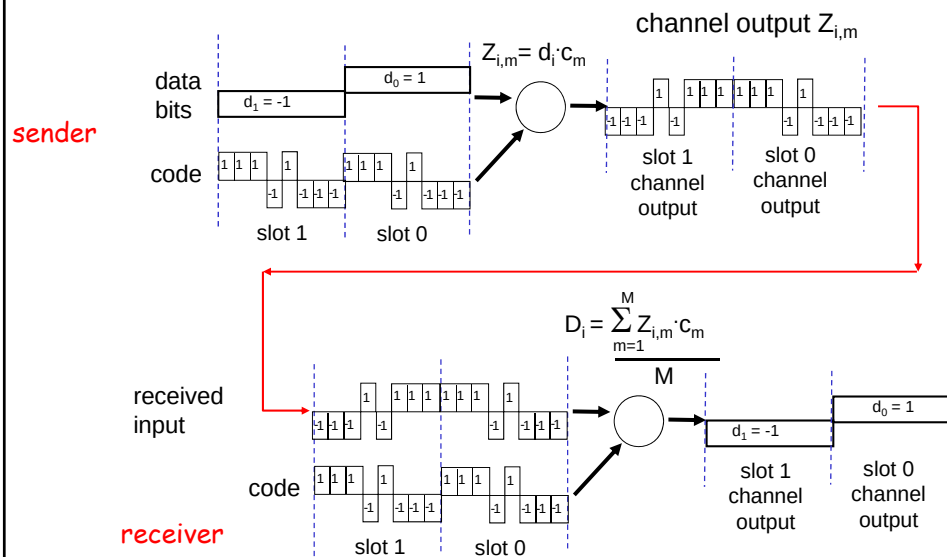
Code Division Multiple Access (CDMA)

- all users share same frequency band, but each user has its own "chipping" sequence (i.e., code) to encode data
- encoded signal** = (original data) X (chipping sequence)
- decoding**: inner-product of encoded signal and chipping sequence
- multiple users transmit simultaneously with minimal interference provided that codes are "orthogonal"

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CDMA Encode/Decode (basic idea)

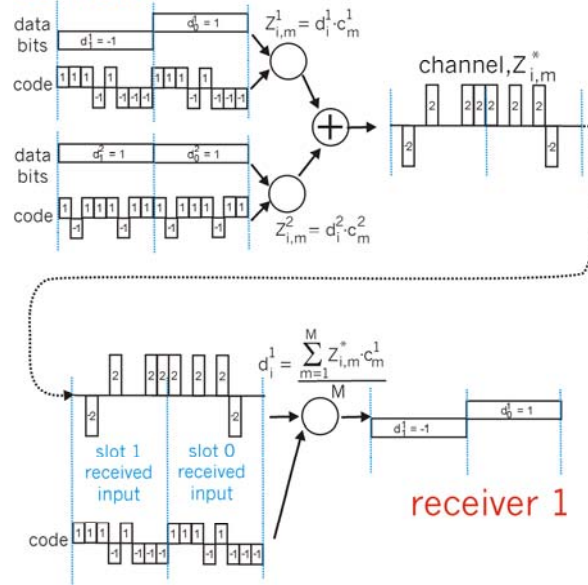


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CDMA: two-sender interference

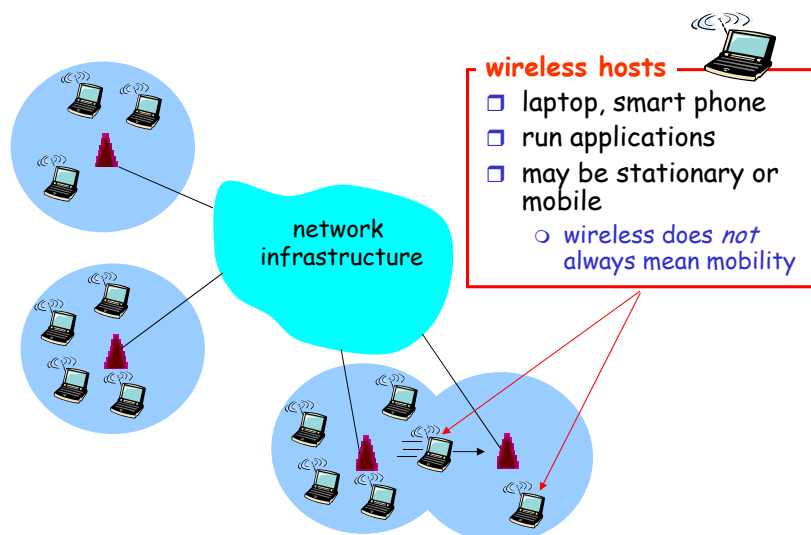
senders



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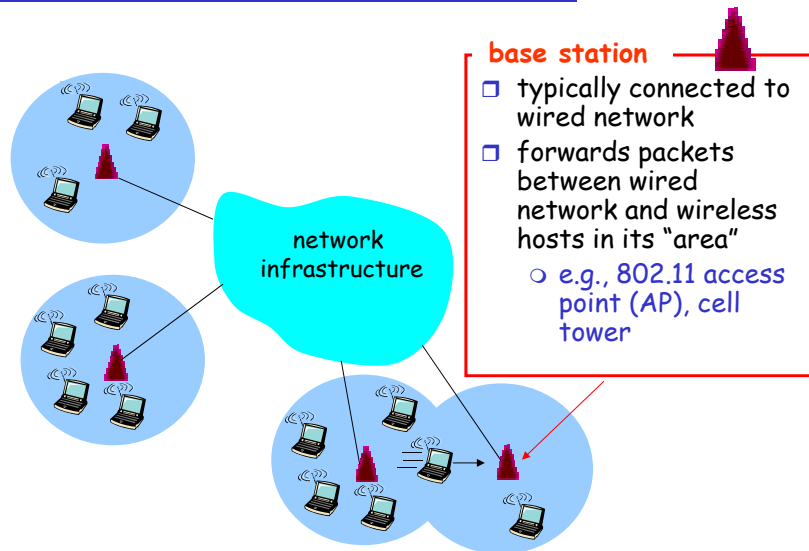
Elements of a wireless network



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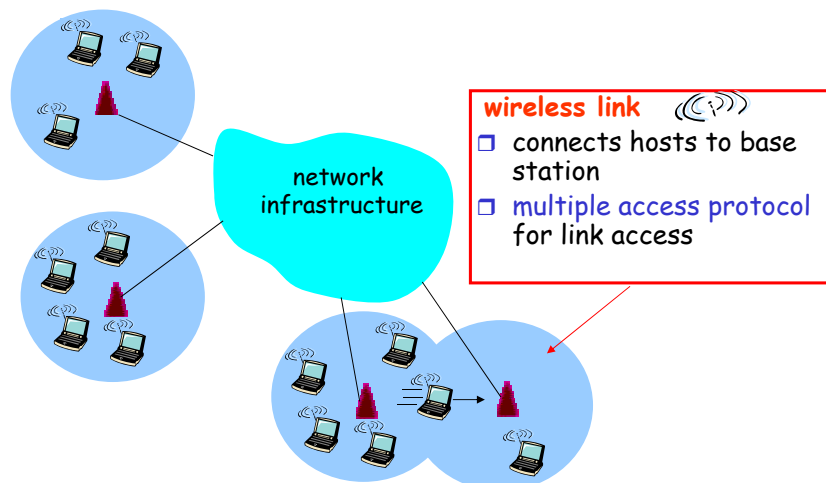
Elements of a wireless network



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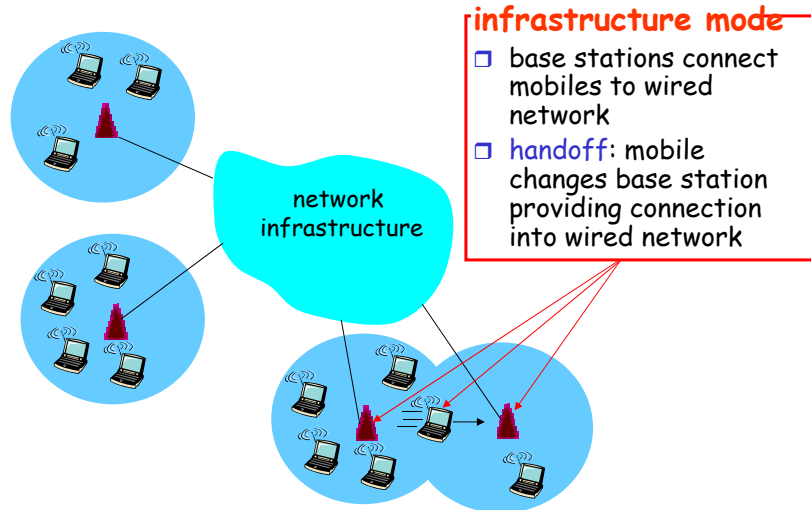
Elements of a wireless network



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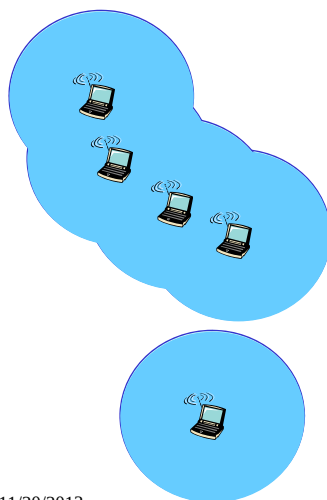
Elements of a wireless network



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Elements of a wireless network



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: route packets among themselves

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Wireless network taxonomy

	single hop	multiple hops
infrastructure mode (e.g., APs)	host connects to base station (WiFi, WiMAX, cellular) which connects to Internet	wireless nodes serve as relays towards Internet, e.g., <i>community mesh net</i>
no infrastructure	no base station, no connection to Internet (Bluetooth, 802.11 ad hoc mode)	no base station, no connection to Internet; wireless nodes serve as relays to reach some nodes: <i>MANET, VANET</i>

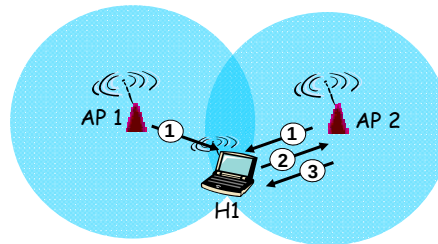
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802.11: channels, association

- frequency band divided into channels

- ❖ AP admin chooses frequency
- ❖ interference possible: channel can be same as that chosen by neighboring AP



- host: must *associate* with an AP

- scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
- selects AP to associate with
 - performs authentication
 - runs DHCP to get IP address in AP's subnet

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Wireless Link Characteristics (1)

Differences from wired link

- **decreased signal strength**: radio signal attenuates more as it propagates (path loss)
- **interference from other sources**: wireless network frequencies (e.g., 2.4 GHz) shared by cordless phones, sensors, etc.; devices (motors) interfere as well
- **multipath propagation**: radio signal reflects off objects, reaching destination at slightly different times

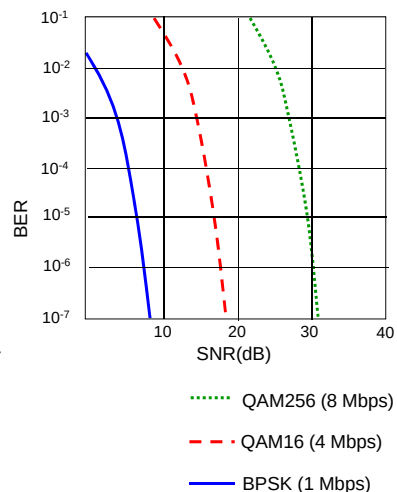
.... make communication across a wireless link more "difficult"

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Wireless Link Characteristics (2)

- SNR: signal-to-noise ratio
 - larger SNR - easier to extract signal from noise
- SNR versus bit error rate (BER) tradeoffs
 - *For a given modulation technique*: increase power $\rightarrow$ increase SNR $\rightarrow$ decrease BER
 - *For a given SNR*: choose modulation technique that meets BER requirement to get highest throughput
- SNR may change with mobility
 - dynamically adapt modulation technique (rate)

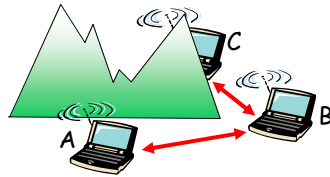


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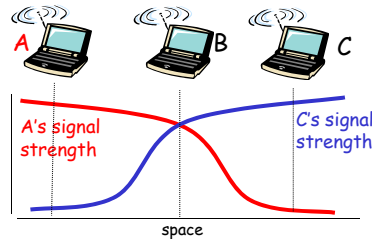
Wireless network characteristics

Multiple wireless senders and receivers create additional problems (**carrier sensing is unreliable**):



Hidden terminal problem

- ☐ B, A hear each other
- ☐ B, C hear each other
- ☐ A, C cannot hear each other
 - > A, C unaware of interference at receiver B



Signal attenuation (fading):

- ☐ B, A hear each other
- ☐ B, C hear each other
- ☐ A, C cannot hear each other, unaware of interference at B

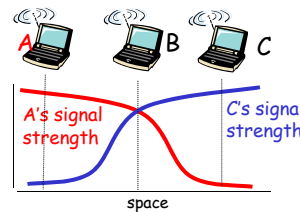
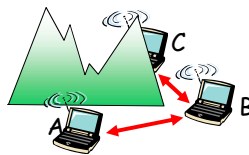
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IEEE 802.11: multiple access

- ☐ 802.11: CSMA - sense before transmitting
- ☐ 802.11: *no collision detection*
 - o difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
 - o can't sense all collisions : hidden terminal, fading

goal: **avoid collisions**, hence CSMA/CA



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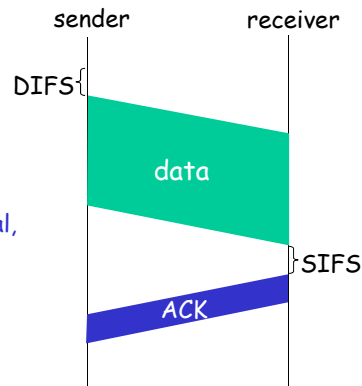
IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

- 1 if channel sensed idle for **DIFS** then
transmit entire frame
- 2 if channel sensed busy then
 - choose random backoff time
 - whenever channel sensed idle, timer counts down
 - *transmit when timer reaches zero*
 - if no ACK, increase random backoff interval, repeat 2

802.11 receiver

- if frame received OK
return ACK after **SIFS**
(ACK needed due to hidden terminal/fading problems)



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Collision Avoidance - idea

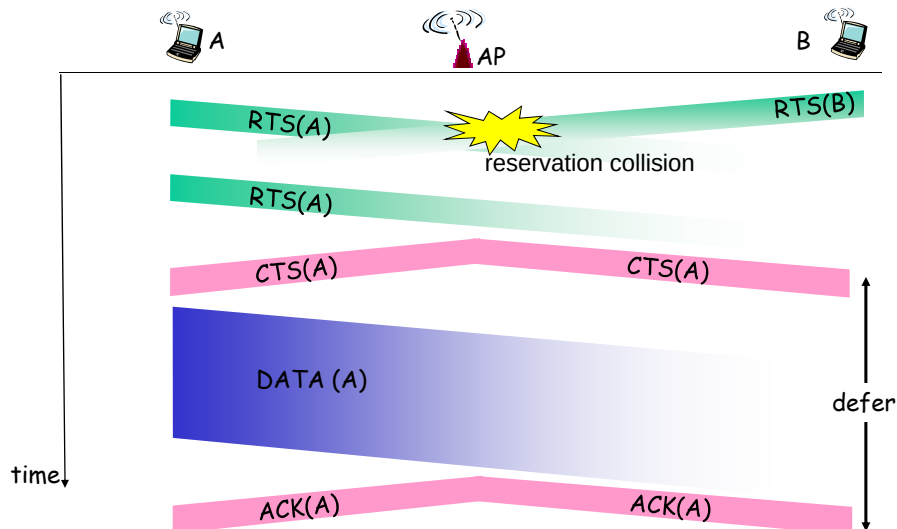
- senders transmit *small* request-to-send (RTS) packets to AP using CSMA
 - RTS packets may collide with each other (but they're **short**)
- AP broadcasts clear-to-send (CTS) in response to RTS
- CTS heard by all nodes
 - the sender selected then transmits data frame
 - other stations defer transmissions

Try to avoid *long* data frame collision
using *small* reservation packets

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RTS-CTS exchange (use is optional)

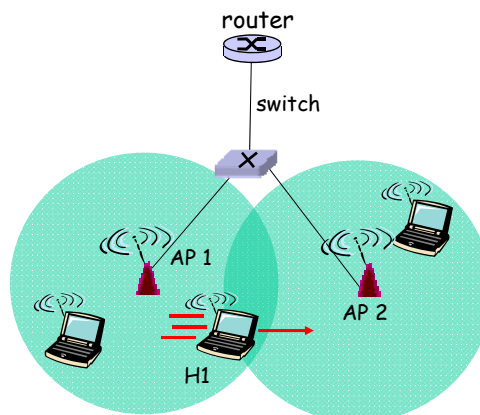


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802.11: mobility within same subnet

- H1 remains in same IP subnet: IP address can remain same
- *switch: which AP is associated with H1?*
 - *self-learning (Ch. 5): switch will see frame from H1 and "remember" which switch port can be used to reach H1*



Suppose the two APs are connected directly to a router.
Mobility ?

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End of our Chapter 6 coverage

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